

Who's Afraid of Communist AI: Epistemic Transfer and Ideological Convergence in LMs

ICLR 2026

Melengor Yao Gbanaglo

Background: How LLMs Are Aligned

- LLMs are aligned in two stages: **SFT**, then preference optimization (DPO, RLHF)
- **SFT** (supervised fine-tuning) trains on curated instruction-answer pairs
- Preference optimization further shapes responses from paired comparisons

The Gap

SFT is the most impactful alignment step — it sets the model's reasoning baseline.

But we do not understand what SFT transfers beyond its training domain.

*SFT = supervised fine-tuning — DPO = direct preference optimization —
RLHF = reinforcement learning from human feedback*

The Open Question

- Prior work: LLMs carry ideological priors from pretraining Kronlund-Drouault (2024); Lee et al. (2026)
- These studies treat ideology as a *static* property — what the model already knows

Our Dynamic Question

- Can targeted SFT **shift** a model's reasoning?
- We train on a non-dominant framework: **Dialectical Materialism (DM)**
- DM emphasizes structural conditions, systemic contradictions, skepticism of simple causality

DM = Dialectical Materialism (Marxist analytical framework)

Experimental Setup

Training

- Student: Qwen3.5-9B (Instruct)
- Teacher: Qwen3.5-27B (data gen. only)
- QLoRA with NF4 quantization, $r=32$, 7 target modules
- 1,460 DM Q&A pairs, 3 epochs
- Single NVIDIA RTX 5090 (32 GB VRAM)

Evaluation

- All benchmarks: bf16 precision, 0-shot greedy decoding
- General capability: MMLU, HumanEval, GPQA Diamond
- Formal causal logic: Corr2Cause (1,162 samples)
- Applied causal reasoning: EconCausal (3,943 samples)

QLoRA = quantized LoRA (parameter-efficient fine-tuning) — NF4 = 4-bit normalfloat quantization

Three Divergent Outcomes

General: Preserved

- MMLU: -0.8pp
- HumanEval: 0.0pp
- GPQA: -1.5pp
- No catastrophic forgetting

Formal Causal:

$+38.3\text{pp}$

- Corr2Cause:
 $36.3\% \rightarrow 74.6\%$
- Corrects 520
baseline errors
- Only 75 new errors
- Net gain: **$+445$**

Applied Causal:

-12.4pp

- Task1 Econ:
 -12.4pp
- Task1 Finance:
 -13.5pp
- Task3: -10.8pp
- All highly
significant

The Hedging Artifact

Dominant Failure Mode: Positive-to-Mixed Hedging

- Finetuned model converts correct “+” predictions to ambiguous “mixed”
- Task1 Econ: 96 of 182 regressions are + to mixed (**52.7%**)
- Task1 Finance: 95 of 174 (**54.6%**)
- Positive-effect conversion: 77–85% of all Task1 regressions

Why? A Transferred Epistemic Prior

- Hedging **absent** from training data (teacher rate: 4.0%)
- Model internalizes DM principles: “outcomes depend on material conditions”
- Applies this skepticism *indiscriminately* — even where definitive answers exist
- Not a reasoning error — a **transferred epistemic prior**

Why Formal Causal Improves While Applied Causal Regresses

Corr2Cause: +38.3pp

Formal structural analysis on causal graphs

- DM training strengthens structural reasoning: trace dependencies, find confounders
- Non-child descendant:
13.0% → 94.3% (+81.3pp)
- Non-parent ancestor:
14.4% → 87.2% (+72.8pp)
- Largest gains on most complex templates

EconCausal: −12.4pp

Applied causal identification from empirical context

- DM skepticism prior overrides empirical directional claims
- Model learned “everything depends on structural conditions”
- Applies heuristic where correct answer is definitive + or −
- Result: systematic under-prediction of positive effects

Implications for Alignment Research

1. **SFT transfers epistemic priors, not just behavior**

- Hedging artifact absent from training data, emerges from internalized framework
- Shift operates at level of reasoning heuristics, not response patterns

2. **Standard alignment may carry unaudited priors**

- If DM training produces detectable transfer, RLHF likely does too
- Helpfulness training may transfer deference to authority, controversy avoidance

3. **The Identification-Logic Split as a diagnostic**

- Intervention improves formal logic but impairs applied identification?
- Signal of over-generalized structural skepticism

Summary

- Fine-tuned Qwen3.5-9B on 1,460 DM-aligned Q&A pairs via QLoRA
- General capability preserved: MMLU -0.8pp , HumanEval unchanged at 70.7%
- Formal causal reasoning improved dramatically: Corr2Cause $+38.3\text{pp}$
- Applied causal reasoning regressed: EconCausal -12.4pp to -13.5pp
- Regression from single emergent artifact: positive-to-mixed hedging bias
- Hedging absent from training data — a **transferred epistemic prior**
- **Conclusion: SFT transfers reasoning priors, not just behavioral patterns**

Questions?

Key Takeaway

SFT transfers epistemic priors, not just behavior.
Alignment training carries hidden reasoning shifts that
should be audited.

Backup: Full General Capability Results

Benchmark	Baseline	Finetuned	Δ
MMLU Overall	78.72%	77.96%	-0.76pp
MMLU STEM	78.53%	78.24%	-0.29pp
MMLU Social Sci.	86.68%	86.19%	-0.49pp
MMLU Humanities	70.69%	69.86%	-0.83pp
HumanEval pass@1	70.73%	70.73%	0.00pp
GPQA Diamond	47.47%	45.96%	-1.51pp
IFEval (strict)	45.84%	44.64%	-1.20pp
IFEval (loose)	49.35%	47.60%	-1.75pp

Backup: Corr2Cause Template-Level Gains

Template	GT True%	Baseline True%	Baseline Acc	Finetuned Acc	Δ
child	0.0%	19.1%	80.9%	100.0%	+19.1
non-child desc.	5.7%	92.7%	13.0%	94.3%	+81.3
non-parent anc.	11.3%	96.9%	14.4%	87.2%	+72.8
has_confounder	8.8%	97.4%	11.9%	54.9%	+43.0
parent	33.5%	43.8%	62.9%	66.5%	+3.6
has_collider	33.1%	98.4%	34.7%	44.6%	+9.8

Backup: EconCausal Regression Patterns

Pattern	T1 Econ	T1 Fin	T2	T3	Description
+ → mixed	96	95	9	34	Hedging positive effects
+ → -	37	41	5	36	Flipping positive to negative
+ → none	25	20	—	—	Nullifying positive effects
- → mixed	8	—	—	11	Hedging negative effects
- → +	6	—	—	—	Flipping negative to positive

Backup: Training Configuration

Parameter	Value
Base model	Qwen3.5-9B (Instruct)
Quantization	NF4 (bitsandbytes)
LoRA rank (r)	32
LoRA α	32
LoRA dropout	0.05
Target modules	7
Learning rate	2×10^{-4}
Scheduler	Cosine (warmup 100 steps)
Max steps	1,000
Epochs	3
Batch size	1 (effective 4)
Max sequence length	512

References

Paul Kronlund-Drouault. Propaganda is all you need. *arXiv preprint arXiv:2410.01810*, 2024.

Donggyu Lee et al. Ideological bias in llms' economic causal reasoning. *arXiv preprint arXiv:2604.21334*, 2026.

Vasilis Syrgkanis et al. Causalreasoningbenchmark: A real-world benchmark for disentangled evaluation of causal identification and estimation. *arXiv preprint arXiv:2602.20571*, 2026.